

Understanding real-world effectiveness of nirsevimab and the characteristics of breakthrough RSV infections: a study protocol of individual-participant-level data meta-analysis (the GEMINI study)

Shaolong Ren ^{1,2} Han Zhang,^{1,2} Shuyu Deng,^{1,2} Harish Nair,^{1,3} You Li ^{1,2,3,4}

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ABSTRACT

Introduction The recent rollout of nirsevimab across multiple countries represents an important advancement in respiratory syncytial virus (RSV) prevention. However, real-world evidence on how its effectiveness varies with infant age and time since administration, and on the characteristics of breakthrough infections, remains limited.

Methods and analysis In this study, we aimed to conduct an international collaborative study to assess the real-world effectiveness of nirsevimab by monthly interval of chronological age and time since administration and to understand the characteristics of breakthrough infections. Both individual-participant-level data (IPD) and aggregated data will be collected from international collaborators identified by systematic review, existing collaborative network and academic conferences. The effectiveness of nirsevimab will be assessed by a test-negative study design, and a two-stage IPD meta-analysis will be conducted for data synthesis; a one-stage IPD meta-analysis will also be conducted among a subset of IPD data with sufficient sample size as an exploratory analysis. We will also describe the epidemiological, clinical and molecular features of breakthrough infections and compare them with infants and young children who were either unvaccinated and infected or vaccinated but not infected. The study will provide detailed insights into the dynamic effectiveness of nirsevimab and the characteristics of breakthrough cases.

Ethics and dissemination The study protocol has been approved by the Ethics Committee of Nanjing Medical University (2025–179). Ethical approval for the contributing studies is expected to be obtained independently by local investigators from their respective ethics committees. Findings will be presented at international conferences and submitted to peer-reviewed journals for publication.

BACKGROUND

Respiratory syncytial virus (RSV) is a leading cause of acute lower respiratory tract infection (ALRI) and a major contributor to infant hospitalisation and mortality worldwide. In 2019, RSV was estimated to cause approximately 12.9 million ALRI episodes,

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Nirsevimab prevents respiratory syncytial virus (RSV) in infants, but real-world effectiveness by infant age and time since administration, as well as breakthrough infection patterns, are not yet fully understood.

WHAT THIS STUDY ADDS

⇒ Pools global individual-level and aggregated-level data to estimate the dynamic effectiveness of nirsevimab and characterise profiles of breakthrough RSV infections in infants and young children.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ By clarifying the dynamic effectiveness of nirsevimab and the profiles of breakthrough cases, this study will help optimise nirsevimab implementation in practice.

2.2 million hospital admissions and 66 300 deaths among infants globally.¹ Nirsevimab is a long-acting monoclonal antibody that targets the prefusion conformation of the RSV F protein, thereby inhibiting viral fusion and preventing infection.² Nirsevimab received initial authorisation in the European Union in November 2022,³ becoming the first immunoprophylactic agent globally approved for the prevention of RSV in the general infant population. Since then, regulatory approval has expanded rapidly; as of June 2025, nirsevimab had been authorised in 54 countries, 18 of which have incorporated it into their national immunisation programmes. Recent data show high uptake in several countries, including Spain (90.1%), Luxembourg (83.7%), France (76.5%), and the United States (51.2%).⁴



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Correspondence to Professor You Li; you.li@njmu.edu.cn